E1:

2 steps

- Leaving group leaves → carbocation
- Base removes β-H → ene

Rate = k[Sub]

Favored:
- Weak base/nucleophile
- polar protic solvent
- Heat
- 3° > 2° >> 1

Rearrangements possible

Zaitsev

Competes w/ SN1

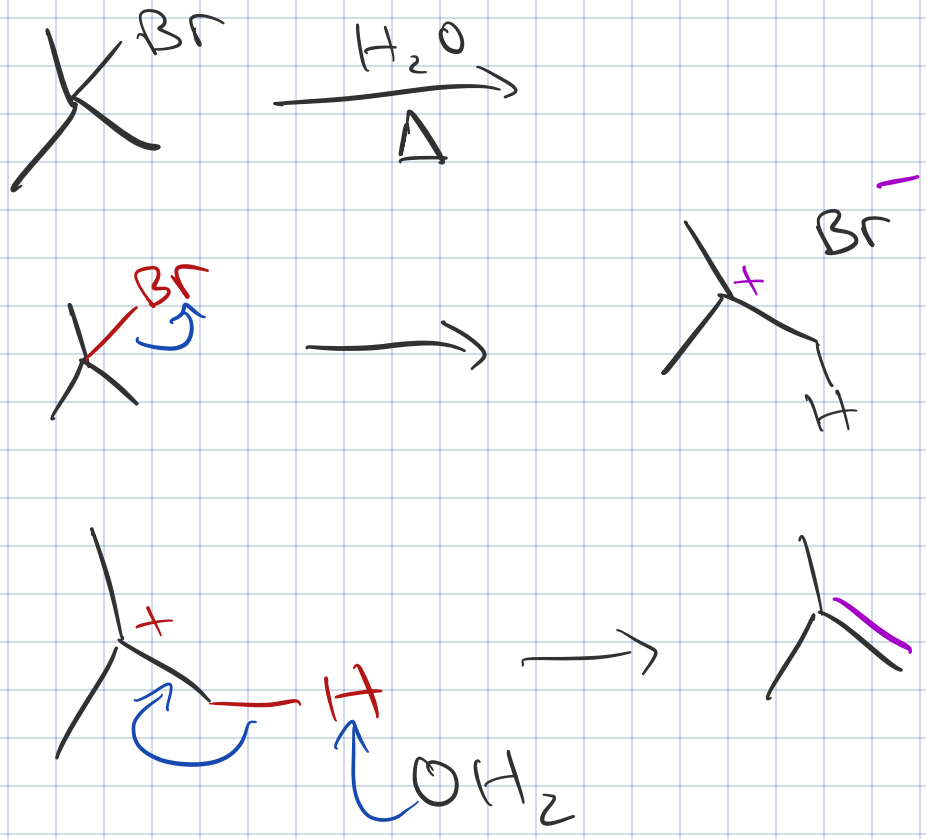
E2:

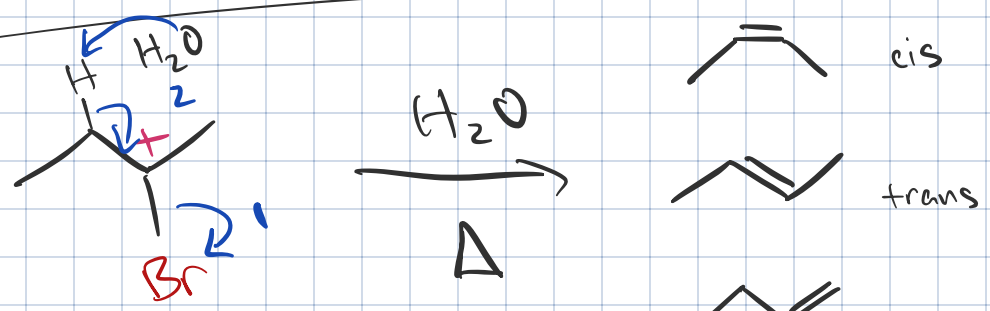
Mech: Concerted (Base attacks β-H and LG leaves)

rate = k[sub][base]

- Strong base
- Aprotic Solvents
- Heat
- 3° & 2° (1° w/ bulky base)
- No rearrangement
- requires anti-periplanar β-H
- Zaitsev major unless steric hindrance
- Competes w/ SN2

E1 Mech Examples



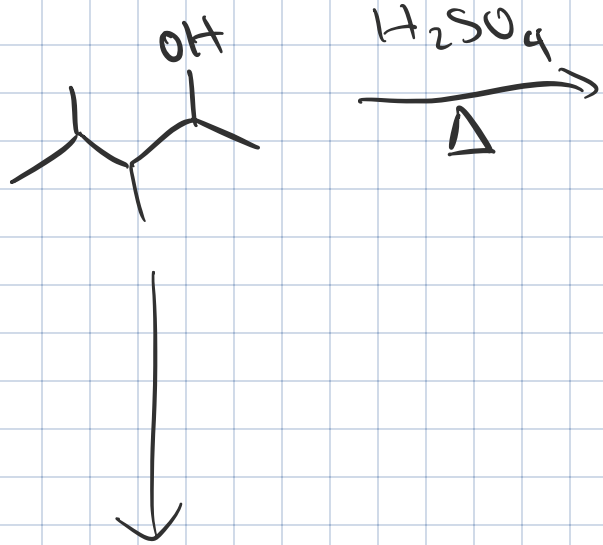


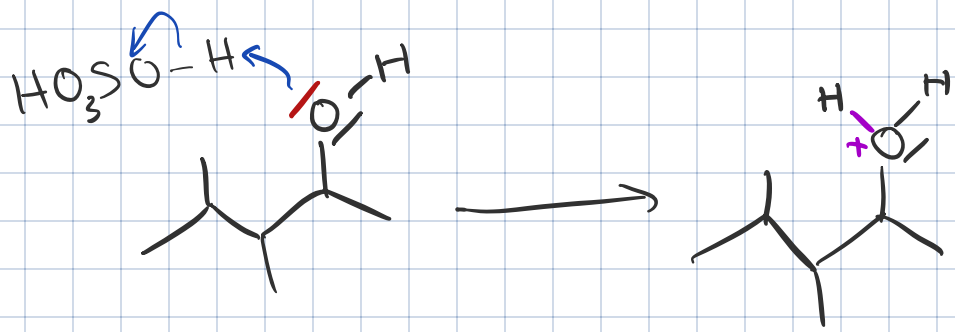
Zait. also minor

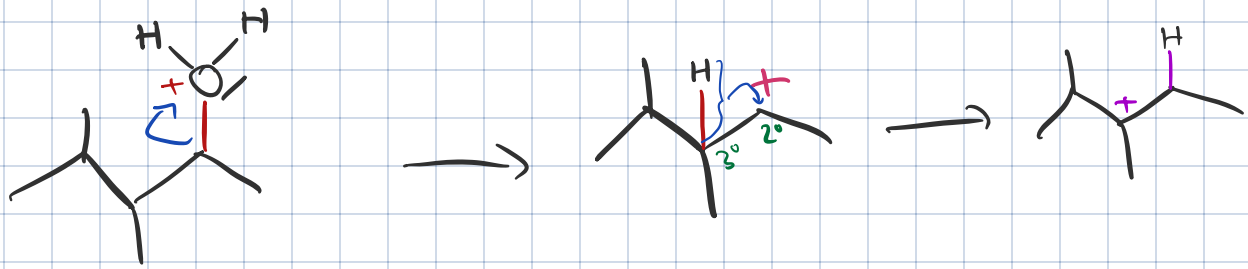
sterics

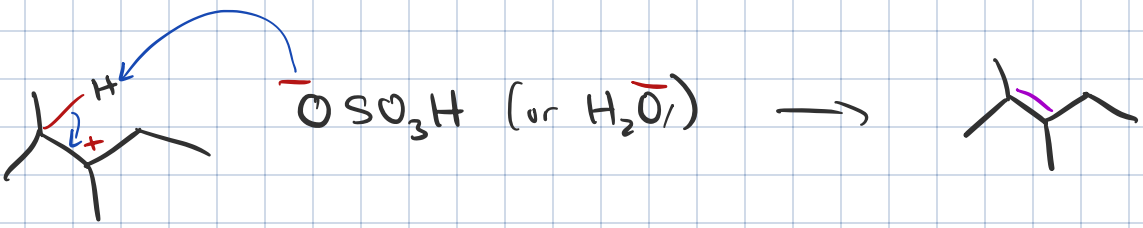
best

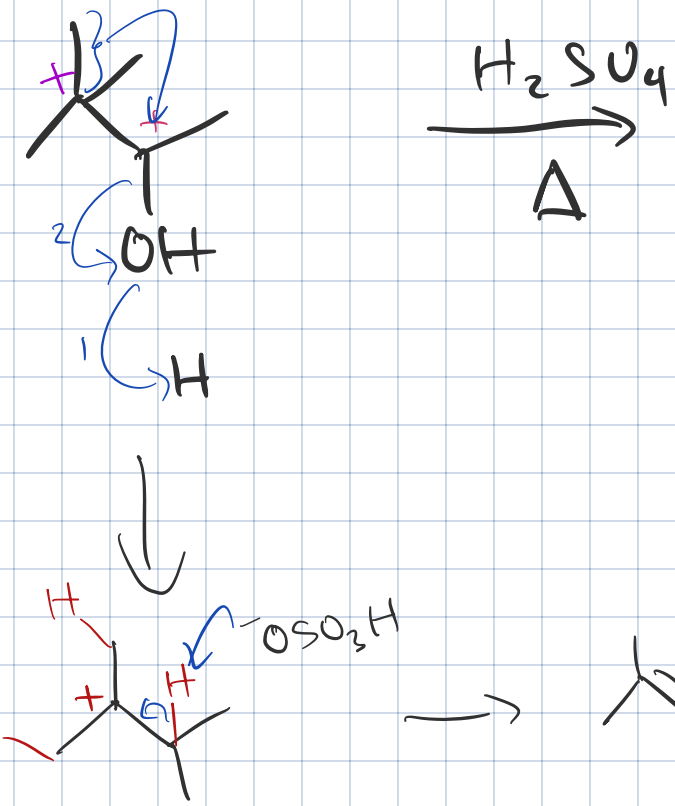
Hoff. minor

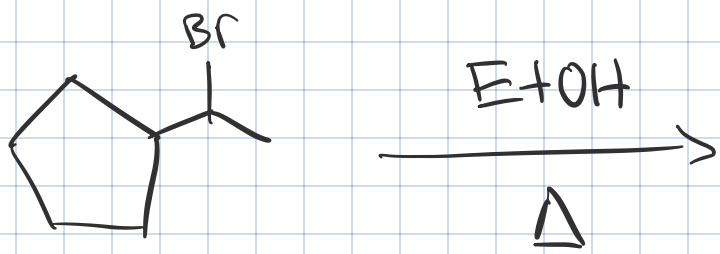


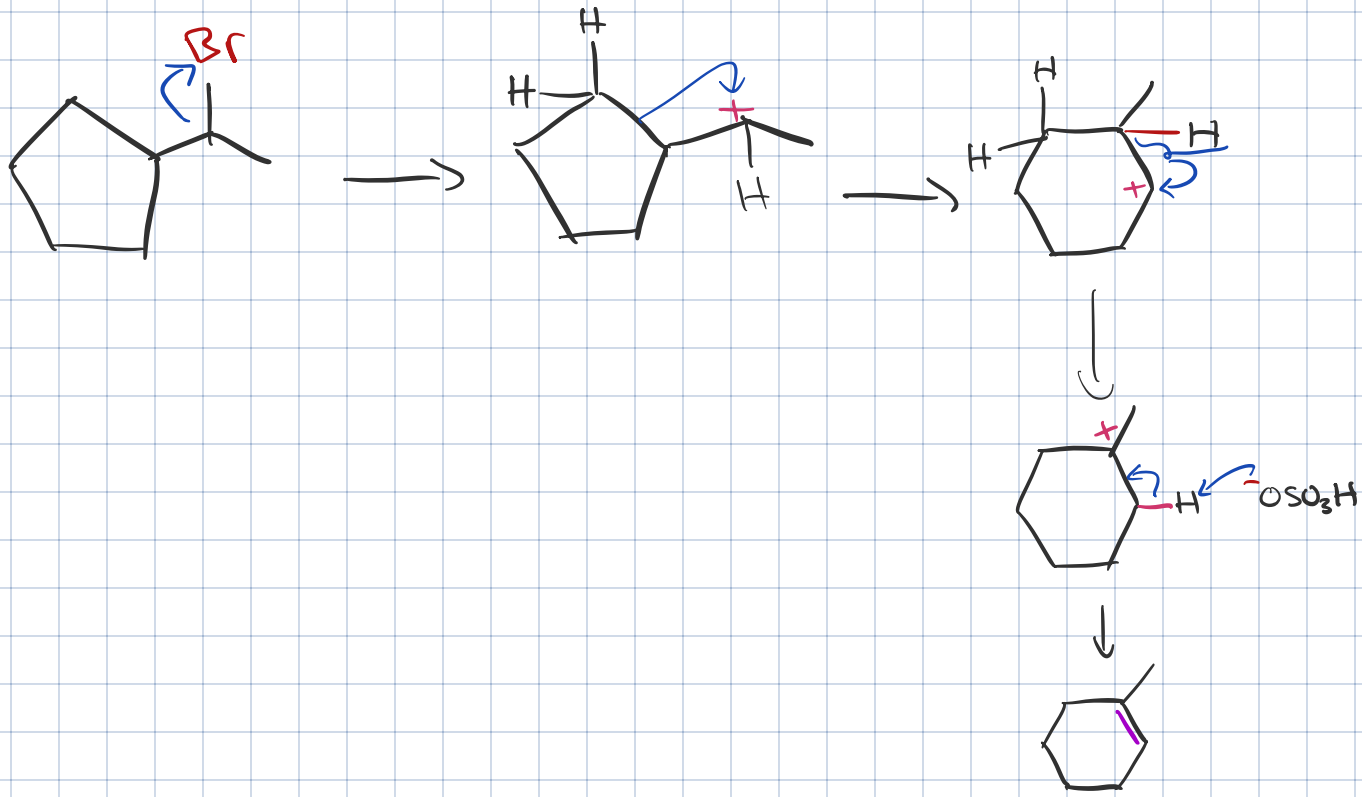

HO_3SO-H
H
H
H


H
H
H
3°
2°
H

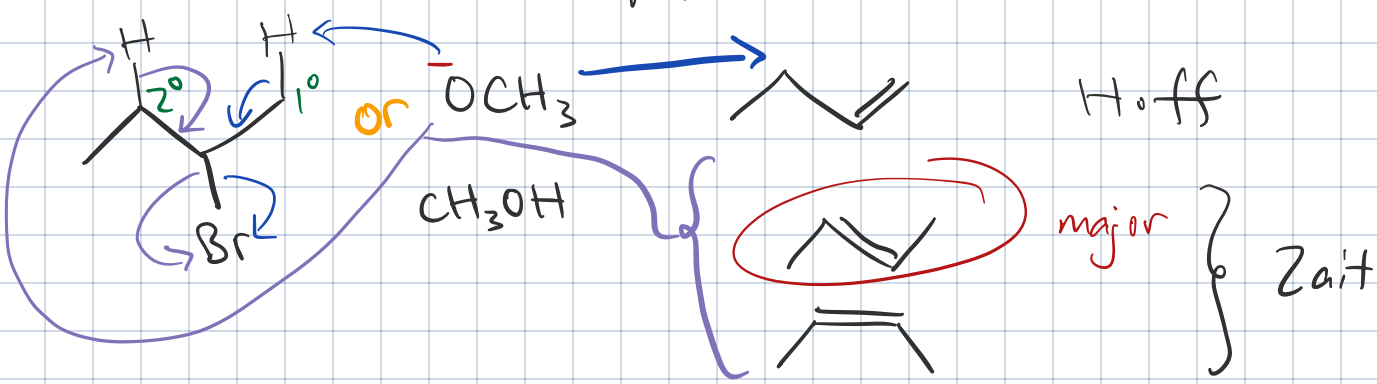

H
$^{-}OSO_3H$ (or H_2O)

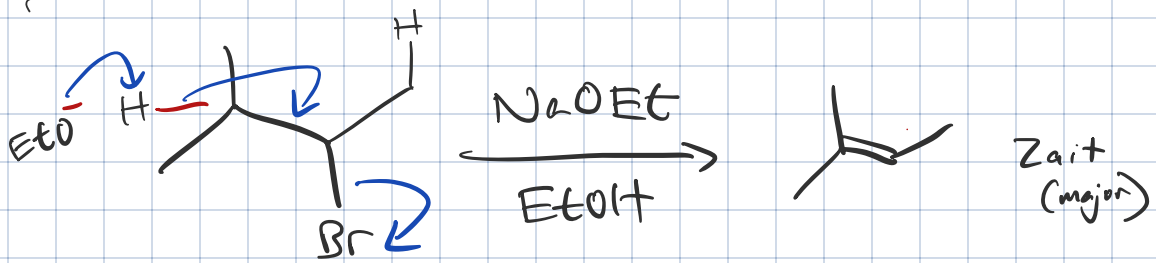

H_2SO_4
Δ
2
OH
1
H
H
H
H
$^{-}OSO_3H$

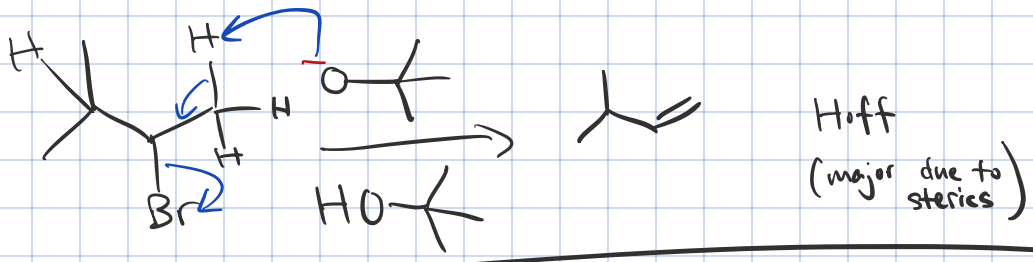


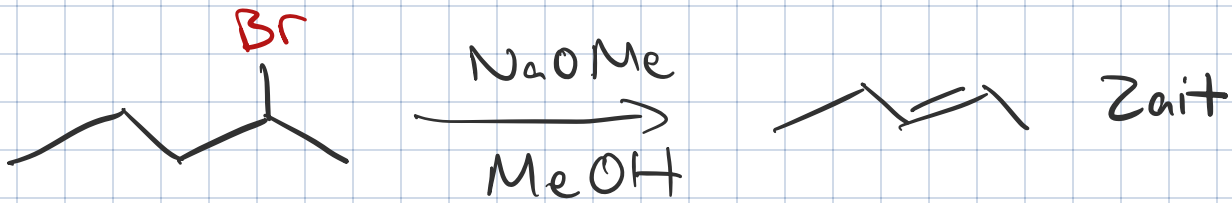


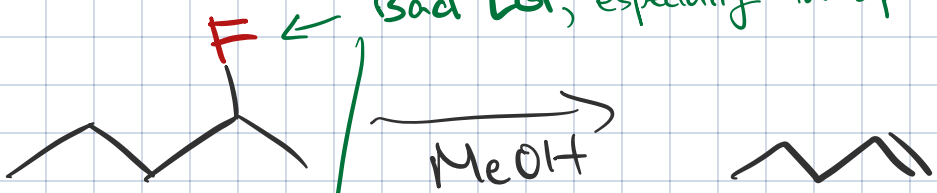
E2 rxn mechanism



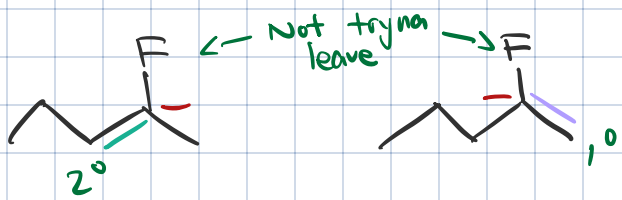






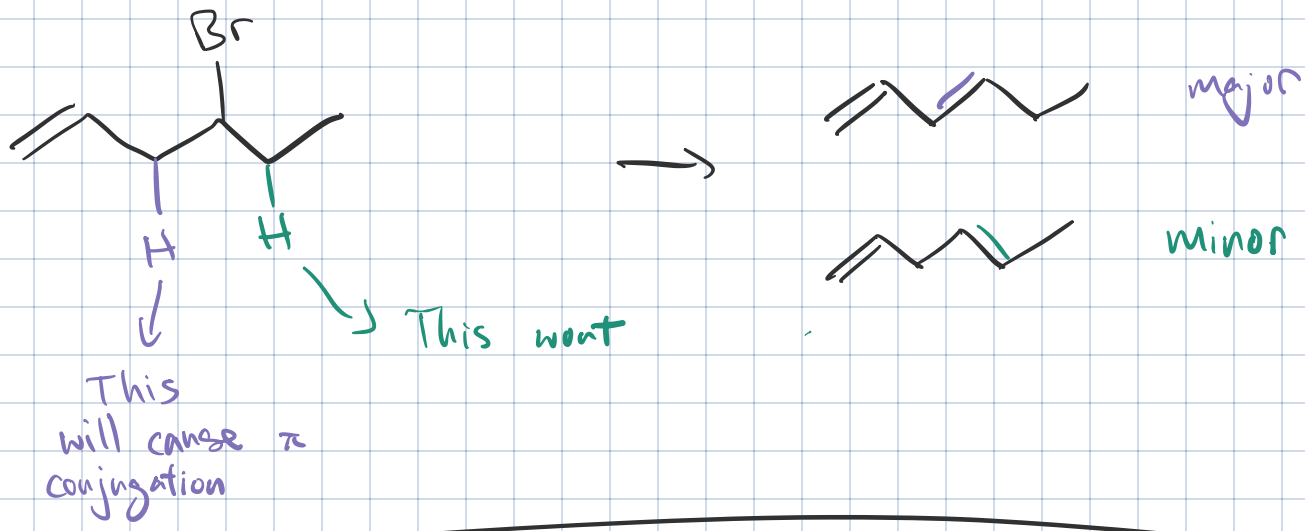


Important: When dealing w/ a bad LG in E2, the transition state dwells on a carbanion:

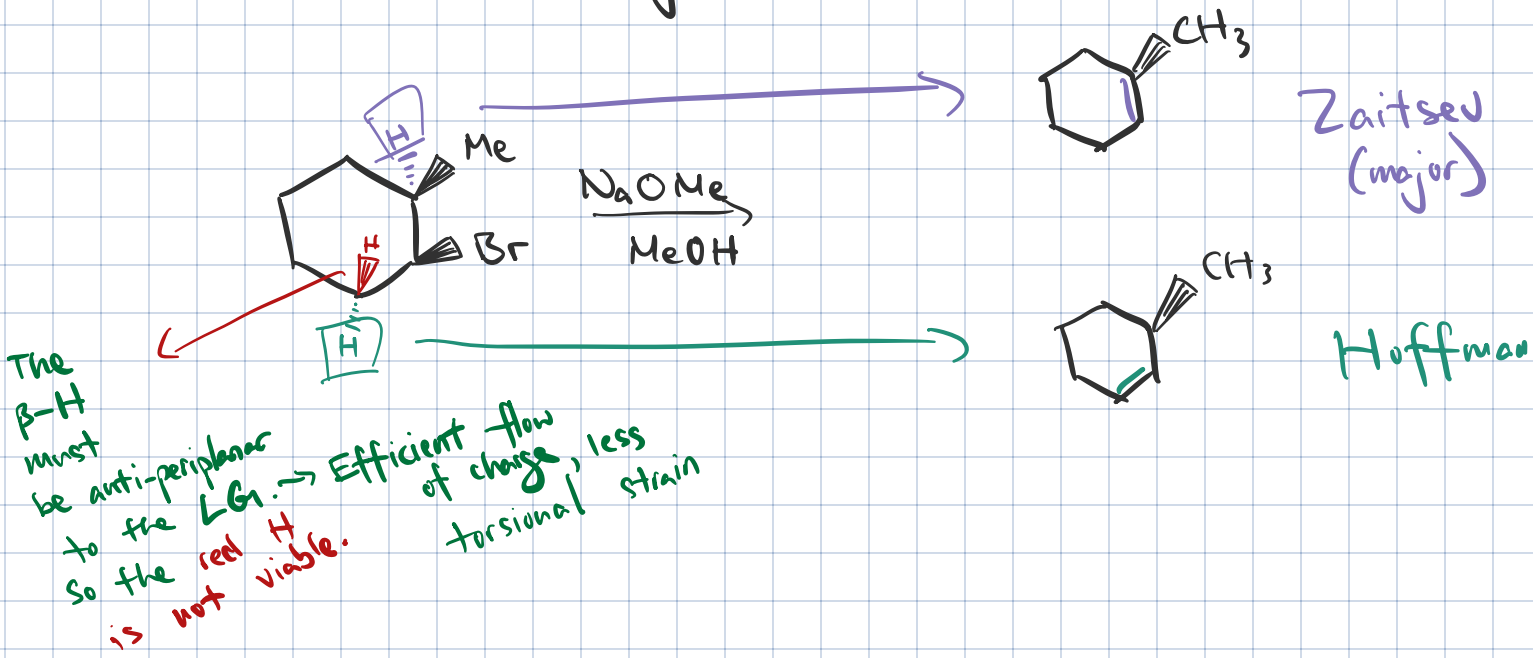


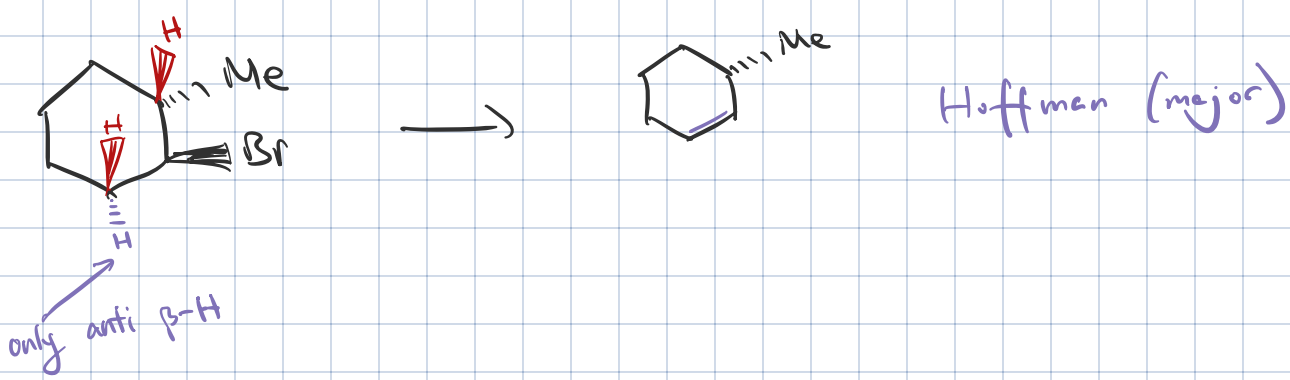
Recall that for carbanions stability you want a less substituted/crowded area, so the transition state on the right above is more accessible, leading to the favorability of the Hoffman

Alkyl fluorides → Hoffman

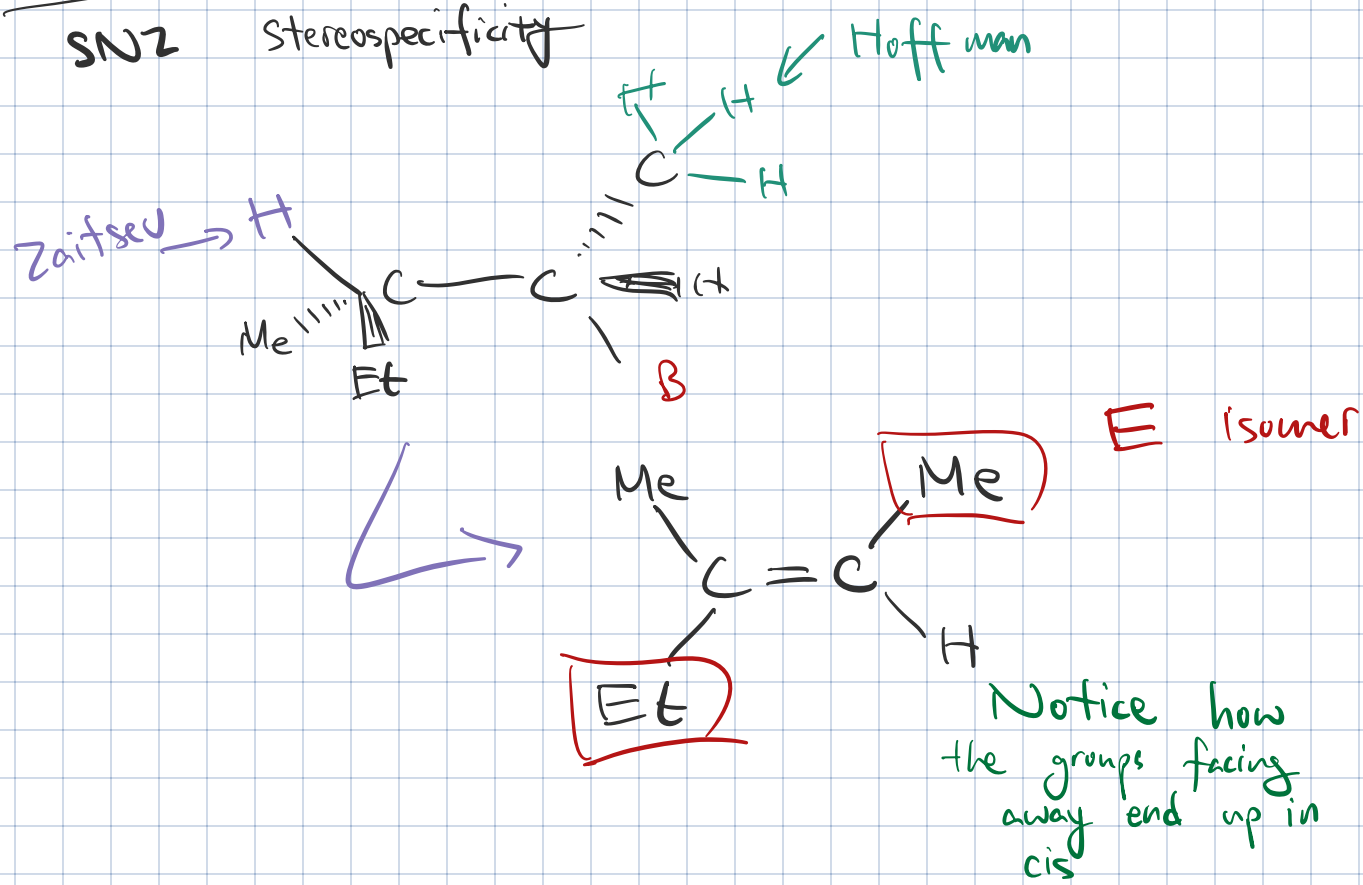


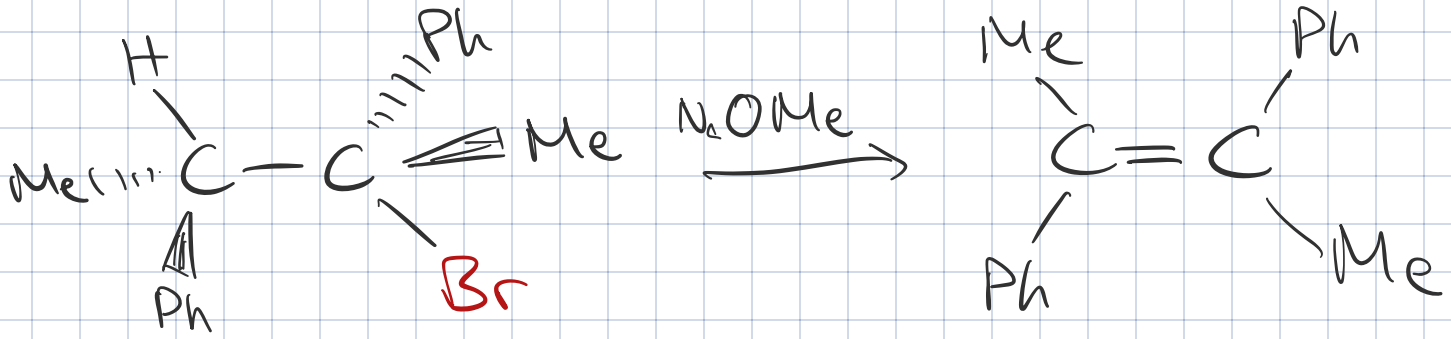
E2 Stereochemistry



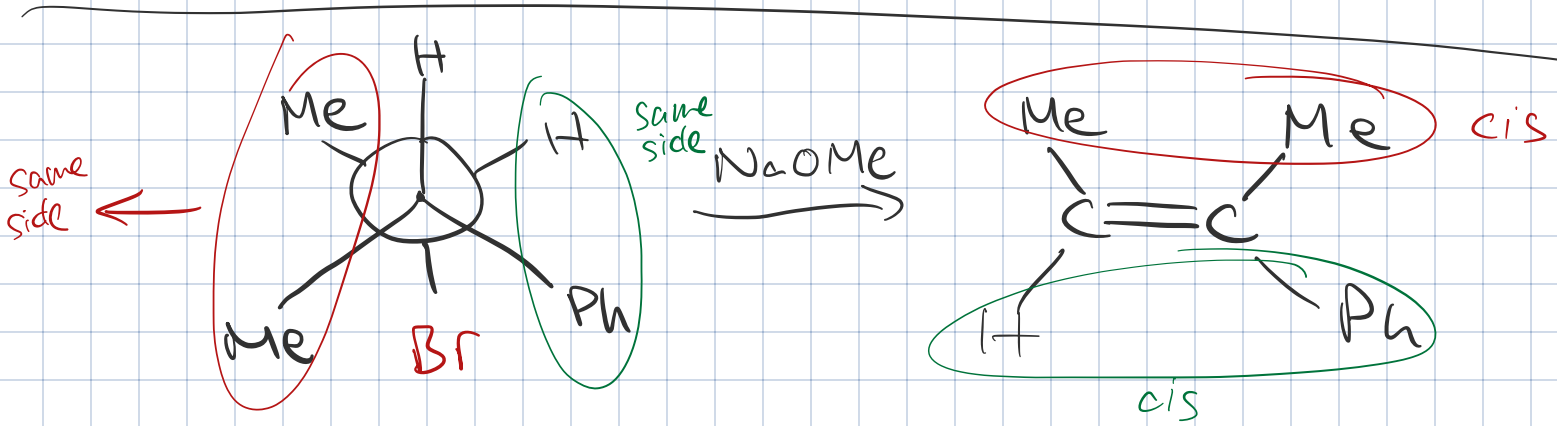


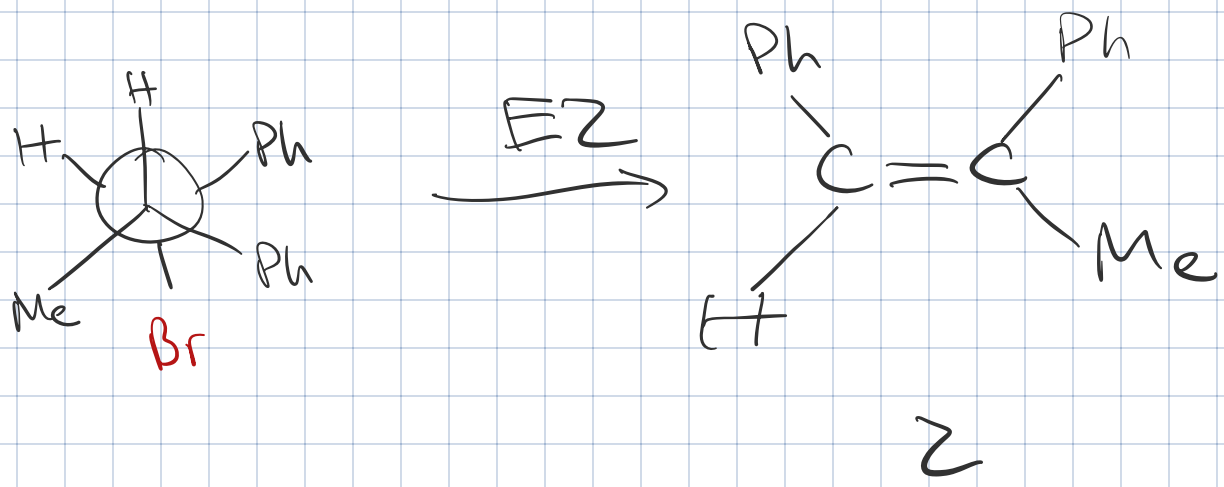
SN2 Stereospecificity





Trans Zaitsev




H
H
Ph
Ph
Me
Br
E2
Ph
Ph
C
C
H
Me
Z